Lecture 13

Values added: contextuality, locality, Bohm

Objectives. *Giving up completeness C means adding physical facts to the state. Those facts cannot simply be noncontextual values of every observable. The Peres–Mermin square proves the obstruction. Bell locality separates into two model-relative conditions. Bohmian mechanics illustrates the cost: additional positions with contextual readings and, in equilibrium, the quantum position distribution preserved by the guiding flow.*

- Prove the square's value contradiction and the noncontextual bound four against the quantum statistic six.
- Prove Bell locality is parameter independence plus outcome independence, keeping setting independence separate.
- Derive the probability current and prove one-dimensional equivariance under the stated smooth-flow hypotheses.

13.1 Which rules should individual values obey?

Expectation values add: $\omega(A+B) = \omega(A) + \omega(B)$, even when A, B do not commute. That does not mean the outcome of reading $A+B$ is a sum of outcomes in one joint reading of A and B. For Pauli matrices σ_x, σ_z, each spectrum is $\{+1, -1\}$, but $(\sigma_x + \sigma_z)^2 = 2\mathbb{1}$, so their sum's spectrum is $\{+\sqrt{2}, -\sqrt{2}\}$. Individual spectral values of the two terms add to $-2, 0, 2$, none of these. A value rule additive on this noncommuting pair is therefore impossible.

Von Neumann's no-hidden-values argument assumes an additivity requirement beyond commuting families; Bell's critique targets that physical assumption (von Neumann pp. 201–202; Bell, ch. 1 p. 4). It does not invalidate the mathematics under its stated assumptions. L3's effect additivity likewise concerns probabilities, not a license to add deterministic eigenvalues of incompatible observables.

A *noncontextual value assignment* here chooses a spectral value $v(A)$ for the listed observables, independent of the commuting family in which it is read, and respects functional relations within each such family. In particular it respects its products. The same observable can belong to different commuting families. Noncontextuality requires its assigned value to remain the same when that family changes; it does not require all the observables to be read together. The usual condition $v(f(A)) = f(v(A))$ implies those rules for finite commuting matrices: assume the value rule also applies to their self-adjoint functions, including the auxiliary matrix below. Simultaneous diagonalisation gives common eigenspace projections (E_r) and a matrix $D = \sum_r d_r E_r$ with distinct d_r. Every member is a function of D, and $v(AB) = (fg)(v(D)) = f(v(D))g(v(D)) = v(A)v(B)$. No rule for a product of noncommuting operators has been introduced.

Self-check 13.1

1. Why does the Pauli example challenge a value assumption but leave linearity of expectations intact?

13.2 The Peres–Mermin square

In this section put $X = \sigma_x$, $Y = \sigma_y$, $Z = \sigma_z$. On two qubits use the nine observables

$$(A_{jk}) = \begin{pmatrix} \mathbb{1}\otimes Z & Z\otimes\mathbb{1} & Z\otimes Z \\ X\otimes\mathbb{1} & \mathbb{1}\otimes X & X\otimes X \\ X\otimes Z & Z\otimes X & Y\otimes Y \end{pmatrix}. \tag{13.1}$$

All are self-adjoint and square to $\mathbb{1}$, with eigenvalues ± 1. Each row or column is one commuting *context*. To check every product,

$$A_{11}A_{12} = A_{13}, \quad A_{21}A_{22} = A_{23}, \quad A_{11}A_{21} = A_{31}, \quad A_{12}A_{22} = A_{32}.$$

The first pairs act on different factors and commute. For the remaining row and column, $XZ = -iY$, $ZX = iY$ give

$$A_{31}A_{32} = (XZ)\otimes(ZX) = Y\otimes Y = A_{33}, \qquad A_{13}A_{23} = (ZX)\otimes(ZX) = -Y\otimes Y = -A_{33}.$$

Reversing either pair gives the same product, so these contexts commute as well; their third member is plus or minus that product. All three row products are $+\mathbb{1}$, the first two column products are $+\mathbb{1}$, and the third column product is $-\mathbb{1}$. The six compatible choices and their product signs appear in Figure 13.1.

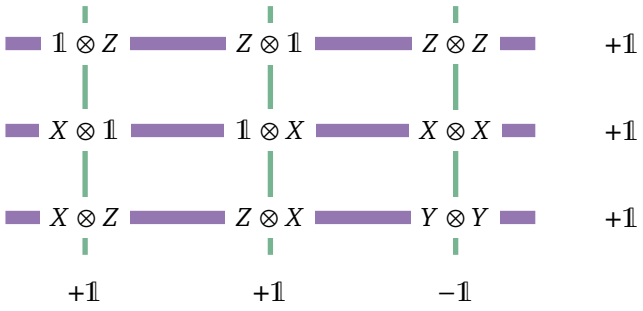


Figure 13.1: Each horizontal row and each vertical column is one compatible context. Product operators appear at the row ends and below the columns; the negative last column prevents one common assignment of nine values.

Theorem 13.1 (Kochen–Specker contradiction on two qubits)**.** *There is no assignment of context-independent values $v(A_{jk}) \in \{\pm 1\}$ respecting the six commuting-triple products of* (13.1).

Proof. Multiplying the three required row-value products gives $+1$. Multiplying the three required column-value products gives -1. Each expression is the product of the same nine scalar values, once each. They cannot differ. No noncommuting operator product was used in this scalar contradiction. ∎

This proof is in dimension four. Dimension three also has Kochen–Specker constructions, including Peres's 33 real rays; Cabello's 18 rays give another dimension-four construction (named only). In finite dimension at least three, the impossibility of assigning zero or one to every projection, adding to one on each orthogonal resolution, also follows from the cited Gleason theorem of L3 and L2's absence of a dispersion-free quantum state; that route depends on the cited theorem. The square is the independent finite proof supplied here. A single qubit escapes this theorem: L3 constructed a two-valued projection assignment, and Bell gives a qubit hidden-value model (ch. 1 pp. 2–3). A statement about all quantum dimensions must retain this distinction.

The same square supplies a statistic. On different runs read each of the six contexts. Let R_j be its row outcome product and C_k its column outcome product, and define the sum of their separately sampled context means

$$\langle \chi \rangle = \langle R_1 \rangle + \langle R_2 \rangle + \langle R_3 \rangle + \langle C_1 \rangle + \langle C_2 \rangle - \langle C_3 \rangle. \tag{13.2}$$

For one noncontextual nine-value assignment write χ for the corresponding scalar sum. Then the product of these six *signed* terms is -1: each value occurs twice and the last term adds one minus sign. At least one term is -1, so $\chi \leq 4$. A probability mixture, with its assignment distribution independent of the context chosen, retains $\langle \chi \rangle \leq 4$. Quantum theory instead makes each signed context-product operator $\mathbb{1}$, giving $\langle \chi \rangle = 6$ for every density operator. The six contexts are separate experimental choices, not a single joint reading of nine noncommuting observables.

The physics

Kirchmair et al., *Nature* **460**, 494 (2009), arXiv:0904.1655, used two trapped calcium ions and sequential compatible readings. Their reported singlet statistic was 5.46 ± 0.04, above four. The test requires stable implementations of each observable across its contexts, and control of compatibility and disturbance. The ideal bound excludes the specified deterministic noncontextual assignments and their mixtures; it is not an assumption-free bound on every noisy stochastic response model. The paper analyses its experimental disturbance conditions.

Self-check 13.2

1. Verify the third row and third column products directly from $XZ = -iY$ and $ZX = iY$.
2. Why can six signed terms have sum at most four under a nine-value assignment?

13.3 Locality is a property of a model

Take settings x, y, outcomes $\alpha, \beta = \pm 1$ and hidden state λ. Allow either countable Λ with preparation weights, or a subset of $\mathbb{R}^d$ with an ordinary probability density, extending the finite case and density remark in QM-I, Lecture 12. The distribution of λ is assumed independent of the later settings. This *setting independence* is a separate premise. A model specifies normalised nonnegative responses $p_\lambda(\alpha, \beta | x, y)$, averaged over that fixed distribution to give the observed probabilities. Conditions may fail on a set of hidden states of probability zero, which changes no averaged prediction; use one common such exception set for the finite CHSH choices. Define marginals by summing the other outcome.

Definition 13.2 (Parameter and outcome independence)**.** *Parameter independence* (PI) says the marginals are independent of the remote setting; denote their common values

by $p_\lambda(\alpha|x,y) = p_\lambda(\alpha|x)$ and $p_\lambda(\beta|x,y) = p_\lambda(\beta|y)$. *Outcome independence* (OI) means

$$p_\lambda(\alpha,\beta|x,y) = p_\lambda(\alpha|x,y)p_\lambda(\beta|x,y).$$

Bell locality means there exist normalised local response distributions $r_\lambda^{(1)}(\alpha|x), r_\lambda^{(2)}(\beta|y)$ such that $p_\lambda(\alpha,\beta|x,y) = r_\lambda^{(1)}(\alpha|x)r_\lambda^{(2)}(\beta|y)$.

Proposition 13.3 (Bell locality $\iff$ PI and OI)**.** *For the specified response probabilities, factorisation holds exactly when PI and OI both hold.*

Proof. Sum the factorisation over either outcome to identify its normalised local factors with the corresponding marginals and obtain PI; substituting those marginals gives OI. Conversely substitute the setting-independent marginals of PI into the OI product to obtain normalised local factors. ∎

The OI definition has no division by zero. When the conditioning marginal is positive, it equivalently says $p_\lambda(\alpha|\beta,x,y) = p_\lambda(\alpha|x,y)$, and similarly on the other side. A zero marginal already forces its associated joint probabilities to be zero by positivity. Setting independence was not proved by this equivalence and is still needed to infer the usual Bell bound. PI compares different remote settings at the same hidden state. OI instead compares outcomes at the same pair of settings and hidden state. Neither comparison is the experimental averaging over hidden states. Jarrett's term "completeness" for OI is not L12's completeness C; we use PI/OI to keep that distinction visible.

For the ordinary quantum singlet model $\lambda = \psi^-$,

$$p(\alpha,\beta|x,y) = \tfrac{1}{4}(1 - \alpha\beta\,\mathbf{a}_x\cdot\mathbf{b}_y), \qquad p(\alpha|x,y) = p(\beta|x,y) = \tfrac{1}{2}.$$

Here $\mathbf{a}_x, \mathbf{b}_y$ are the two measurement axes of QM-I. Thus PI holds. At equal axes $p(+,+) = 0$, whereas the product of the marginals is $1/4$; OI fails. This is compatible with QM-I's proved no-signalling statement: the unconditional local marginal is unchanged by the remote choice. OI concerns conditioning on the remote outcome.

Recall the CHSH bound from QM-I, Lecture 12. For two settings on each side, define the observed correlations by

$$\mathcal{E}_{xy} = \sum_{\alpha,\beta=\pm1} \alpha\beta\, p(\alpha,\beta|x,y), \qquad |\mathcal{E}_{00} + \mathcal{E}_{01} + \mathcal{E}_{10} - \mathcal{E}_{11}| \le 2.$$

Bell locality and setting independence imply the displayed bound. The singlet has correlations $\mathcal{E}_{xy} = -\mathbf{a}_x\cdot\mathbf{b}_y$. For orthogonal unit coordinate axes, choose

$$\mathbf{a}_0 = \mathbf{e}_z, \quad \mathbf{a}_1 = \mathbf{e}_x, \qquad \mathbf{b}_0 = (\mathbf{e}_z + \mathbf{e}_x)/\sqrt{2}, \quad \mathbf{b}_1 = (\mathbf{e}_z - \mathbf{e}_x)/\sqrt{2}.$$

The four correlations are $-1/\sqrt{2}, -1/\sqrt{2}, -1/\sqrt{2}, +1/\sqrt{2}$, in that order. The absolute CHSH combination is therefore $2\sqrt{2}$.

A deterministic model may have outputs $A(x,y,\lambda), B(x,y,\lambda)$ depending on both settings. Its joint point distribution equals the product of its two point marginals, so OI holds automatically. If it also obeyed PI and setting independence, it would satisfy the CHSH bound proved in QM-I. Its pointwise bound averages unchanged over a convergent countable preparation sum, since the summands are bounded and the nonnegative weights sum to one. Hence a deterministic model reproducing the singlet violation must violate PI under those premises. Its averaged marginals can nevertheless be nonsignalling: PI conditions on a finer model state than the experimental average. "No Bell-local model" is not a theorem that one can send a controllable faster-than-light signal.

Self-check 13.3

1. At equal singlet axes, compare $p(+|\beta = +)$ with $p(+)$; which condition fails?
2. Which additional premise about the hidden-state distribution enters the CHSH application?

13.4 Bohm's additional positions and their current

For the spinless scalar theory, a wavefunction of N particles obeys

$$i\hbar\partial_t\psi = -\sum_k \frac{\hbar^2}{2m_k}\Delta_k\psi + V\psi, \qquad V \text{ real}, \quad m_k > 0.$$

It is accompanied by an actual configuration $\mathbf{q}(t)$, beyond the wavefunction. Where $\psi \neq 0$, the guiding equation is

$$\dot{\mathbf{q}}_k(t) = \frac{\hbar}{m_k}\operatorname{Im}\frac{\nabla_k\psi}{\psi}(t, \mathbf{q}(t)). \tag{13.3}$$

For smooth solutions, conjugating the Schrödinger equation and subtracting cancels the real potential and gives

$$\partial_t|\psi|^2 = -\sum_k \frac{\hbar}{m_k}\operatorname{Im}(\bar{\psi}\Delta_k\psi) = -\sum_k \nabla_k \cdot \mathbf{j}_k, \qquad \mathbf{j}_k = \frac{\hbar}{m_k}\operatorname{Im}(\bar{\psi}\nabla_k\psi).$$

The gradient-square term in the divergence is real and contributes no imaginary part. Therefore the velocity is $\mathbf{v}_k = \mathbf{j}_k/|\psi|^2$. This continuity equation suggests preservation of the $|\psi|^2$ distribution, called *equivariance*. We prove it on the line with explicit flow assumptions, rather than inferring global existence from a local differential identity. Electromagnetic vector potentials require the gauge-covariant current and are not included in this displayed model.

Theorem 13.4 (One-dimensional equivariance, conditional flow form)**.** *Let $\psi_s(x)$ be a normalised Schrödinger solution on $0 \le s \le t$, smooth enough for the continuity equation, with $\psi_s(x) \neq 0$ everywhere and jointly continuous $\partial_x v(s,x)$. Suppose there exists a family $X_s(a)$, continuous jointly in (s,a), solving $\dot{X}_s(a) = v_s(X_s(a))$, $X_0(a) = a$, throughout this slab for every $a \in \mathbb{R}$. Then a position initially distributed with density $|\psi_0|^2$ has density $|\psi_s|^2$ at each time s.*

Proof. We first show that the flow preserves the order of positions. Then the continuity equation preserves mass between two moving trajectories. Normalisation and strict positivity finally show that no final interval is omitted by the flow.

Trajectories cannot cross. On a compact tube up to a putative first meeting, $|\partial_x v| \le M$. Their positive gap $D_s = X_s(b) - X_s(a)$ obeys $\dot{D}_s \ge -MD_s$, so differentiation of $e^{Ms}D_s$ yields $D_s \ge (b-a)e^{-Ms} > 0$. Thus the flow is strictly ordered. For finite $a < b$, ordinary differentiation of the moving interval gives

$$\frac{d}{ds}\int_{X_s(a)}^{X_s(b)} |\psi_s(x)|^2 dx = -j_s(X_s(b)) + j_s(X_s(a)) \\ + |\psi_s(X_s(b))|^2 v_s(X_s(b)) - |\psi_s(X_s(a))|^2 v_s(X_s(a)) = 0.$$

The endpoint terms cancel because $j = |\psi|^2 v$. Hence every such image interval carries exactly the initial mass between a, b. Let $a \to -\infty$, $b \to +\infty$. That initial mass tends to one, as does the total mass of the normalised ψ_s. The ordered image endpoints must therefore

reach $-\infty, +\infty$: otherwise a nonempty excluded interval would have positive mass, since $|\psi_s|^2$ is strictly positive and continuous. The continuous ordered map is onto. For any final interval choose its endpoints' unique preimages. The moving-interval identity makes its probability under the transported initial position equal its $|\psi_s|^2$ integral. These interval probabilities determine the position distribution, proving equivariance. ∎

Global flow existence, nodes, singularities and the N-particle theorem are *cited*, not proved by the two-line continuity equation: Dürr–Teufel pp. 152–153; Tumulka, Theorem 1.3 pp. 28–29; Berndl et al., *CMP* **173**, 647–674 (1995), arXiv:quant-ph/9503013. Those results have regularity and potential hypotheses and generally assert almost-sure existence under quantum equilibrium, not existence for every arbitrary wavefunction or starting point. A stationary oscillator ground state gives a simple limiting case. Its spatial wavefunction is real and nowhere zero. Time evolution adds only a common phase, which cancels from the ratio of the spatial derivative to the wavefunction in the guiding equation. The velocity therefore vanishes. Each actual position stays fixed, and the stationary quantum position density is preserved. This does not say that every Gaussian packet has zero velocity.

The freely spreading Gaussian packet of QM-II supplies a nowhere-zero example with a linear velocity and global flow; its trajectory calculation is an example for Sheet 13, not an additional existence theorem assumed here.

Self-check 13.4

1. Which terms cancel when differentiating the mass between two trajectories?
2. Why are normalisation, positivity and the onto-flow step needed after the finite-interval identity?

13.5 What the theorems cost this theory

The additional position is an actual physical fact, so the wavefunction alone is not complete C in L12's sense. Quantum equilibrium specifies the initial configuration density $|\psi|^2$, and equivariance preserves it under the proved conditions. Extending agreement to general recorded observations requires modelling the apparatus configuration and its coupling; the scalar one-dimensional proof alone is not that general measurement theorem.

For a concrete pointer description, let x be the system coordinate and y a pointer coordinate. Suppose a specified apparatus interaction has produced the normalised outgoing wavefunction

$$\Psi(x,y) = a\psi_0(x)\varphi_0(y) + b\psi_1(x)\varphi_1(y), \qquad |a|^2 + |b|^2 = 1.$$

Here the system packets and pointer packets are unit vectors on the line. The pointer packets are smooth and have disjoint position supports. The actual apparatus has one pointer coordinate q_M. If it lies in the support of φ_0 with $\varphi_0(q_M) \neq 0$ and $a \neq 0$, the conditional system wavefunction $\Psi(\cdot, q_M)$ is proportional to ψ_0. The second term vanishes at that actual pointer position. A position in the other packet gives the other conditional branch.

In quantum equilibrium, integrating the joint position density over the first pointer support gives

$$\int_{\operatorname{supp}\varphi_0}\int_{\mathbb{R}} |\Psi(x,y)|^2 dx\,dy = |a|^2 \int_{\mathbb{R}} |\psi_0(x)|^2 dx \int_{\operatorname{supp}\varphi_0} |\varphi_0(y)|^2 dy = |a|^2.$$

Integrating over the second gives $|b|^2$. Thus the additional configuration identifies the actual pointer record, while equilibrium supplies its statistics. The full wavefunction still contains both terms and obeys the Schrödinger equation. This conditional branch description is effective while the pointer packets remain separated; it is not a physical GRW collapse. Real overlapping tails make the separation and the conditional branch identification approximate. The example explains the record mechanism for this assumed outgoing state, rather than proving that every apparatus produces it.

In Bohmian spin models, spin belongs to the spinor wavefunction and the apparatus interaction, not to an extra little vector attached to each particle. The velocity uses $(\hbar/m)\,\mathrm{Im}(\psi^\dagger\nabla\psi)/(\psi^\dagger\psi)$ in the corresponding model without a vector potential (cited: Dürr–Teufel, §8.4 pp. 158–161). A Stern–Gerlach outcome is a position record produced by that interaction. Different contexts can associate different outcomes to the same operator while preserving its quantum statistics. For two qubits, contextuality is required by the square's premises. For a single spin one half it is a feature of the Bohmian apparatus model, not forced by a nonexistent qubit KS contradiction. An optional cited illustration of the single-spin mechanism is Tumulka's inverted Stern–Gerlach example (§2.3.9 pp. 79–80, *cited*). It reverses the magnet and exchanges the output labels to preserve the spin calibration. The two arrangements can then assign opposite calibrated outcomes to the same initial wavefunction and configuration. This account uses its spinor apparatus model and approximately separated packets, with spatial/velocity assumptions and small tails neglected; it is not derived from the scalar flow theorem above.

For Bell's experiment the deterministic fine state includes wavefunction and configurations of both systems *and the relevant apparatus*. Settings remain independent experimental inputs in the Bell comparison. Averaging unknown apparatus positions without including them in λ need not leave a deterministic response model; whether it does depends on the actual response functions. At the fully specified level OI holds, and reproducing the singlet violation forces PI to fail, while equilibrium averaged statistics remain nonsignalling. This is the precise model-relative locality cost, rather than a claim of controllable signalling or a premise shared by every proposed theory.

Pusey–Barrett–Rudolph's theorem is named only: under preparation independence for arbitrarily many independent copies and reproduction of the relevant quantum predictions, an ontological model (a hidden-state preparation/response model of the kind defined above) assigns non-proportional pure states non-overlapping hidden-state preparation distributions (ψ-onticity): in the countable case their overlap $\sum_\lambda \min\{\mu_\psi(\lambda), \mu_\varphi(\lambda)\}$ is zero, and in the density case the corresponding integral is zero (Tumulka, Theorem 5.6 pp. 212–213). It is not an assumption-free verdict about what a wavefunction must mean. The next lecture turns to changing U and changing D, on the same explicit-premise basis.

What does adding values cost?

Noncontextual values respecting commuting quantum products fail on the Peres–Mermin square. Bell locality is PI plus OI, with independent settings also needed in the Bell argument. Bohm adds configurations, uses contextual readings and reproduces equilibrium position distributions by equivariance; its deterministic Bell model violates PI while retaining nonsignalling averages. These statements specify the assumptions changed, not a verdict choosing one response to the measurement problem.

Summary. The square gives a finite state-independent contradiction and a statistic six

against four. PI/OI separate the two parts of a factorised response without dividing by zero. The guiding current supplies a conditional one-dimensional equivariance proof with a genuine global distribution conclusion. Nodes and general global existence remain cited. Context and the full apparatus state matter when connecting those positions to readings.

Exercises. Try Sheet 13 in the separate *Exercise sheets* booklet (English).

Self-check answers.

13.1 Expectations are ensemble means; assigning individual spectral values does not inherit additivity for incompatible readings.

13.2 1. The products are $Y \otimes Y$ and $-Y \otimes Y$, respectively, before multiplying the third member. 2. Their product is -1, so at least one is negative; five plus ones and one minus one give at most four.

13.3 1. Zero versus 1/2; OI fails, while PI holds. 2. Setting independence of the hidden-state preparation distribution.

13.4 1. The continuity-equation flux and moving-boundary transport at each endpoint. 2. They exclude omitted final intervals and let every final interval be reached from an initial one.

Sources. Peres, *Quantum Theory: Concepts and Methods*, equation (7.1) p. 189, pp. 196–200; Bell, *Speakable and Unspeakable*, 2nd ed., ch. 1 pp. 2–4; Mermin, *PRL* **65**, 3373 (1990); Cabello, Estebaranz and García-Alcaine, *Phys. Lett. A* **212**, 183 (1996); Jarrett, *Noûs* **18**, 569–589 (1984), pp. 578–583; Dürr–Teufel, *Bohmian Mechanics*, pp. 152–153, 158–161; Tumulka, pp. 28–29, 79–80, 212–213. Experimental and global-existence primary sources are identified in the text.